

GAOUA, Burkina Faso

WMO: 655220

Lat: 10.333N

Lon: 3.183W

Elev: 1099

StdP: 14.12

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	59.6	61.6	18.3	14.4	88.2	22.0	17.2	87.6	14.5	87.2	14.0	86.5	0.2	20	0.328

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	21.7	101.1	69.4	99.7	69.6	98.4	69.7	79.8	89.3	79.1	88.9	78.4	88.3	5.1	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.7	150.0	83.4	76.9	145.9	82.9	76.2	142.5	82.5	44.3	89.0	43.5	88.9	42.9	88.4	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.3	12.8	11.5	54.7	104.1	3.0	1.1	52.5	104.9	50.8	105.6	49.1	106.2	47.0	107.1		
(5)			46.1	82.2	1.5	1.2	45.0	83.0	44.2	83.7	43.3	84.3	42.3	85.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.1	77.7	83.0	87.9	88.0	85.5	82.4	79.9	78.8	79.8	81.9	82.0	78.0
	DBStd	4.40	2.88	4.05	3.09	3.28	3.10	2.88	2.32	1.97	2.28	2.21	2.29	2.67
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11704	859	924	1175	1140	1099	973	927	894	894	990	960	869
	CDD65	6229	394	504	710	690	634	523	462	429	444	525	510	404
	CDH74	73450	5002	6823	10264	9799	8236	5903	4207	3396	3744	5238	5876	4962
(13)	CDH80	37310	2731	4000	6240	5743	4251	2433	1357	911	1274	2357	3273	2740
(14)	Wind	WSAvg	4.0	3.5	3.9	4.4	5.6	5.4	4.6	4.0	2.6	2.7	2.4	3.3
(15) Precipitation	PrecAvg	42.0	0.1	0.2	1.1	2.9	4.6	5.2	7.3	9.0	7.3	3.4	0.4	0.2
	PrecMax	61.6	3.7	2.3	4.1	8.7	9.6	10.3	16.2	22.8	14.0	12.2	3.0	2.9
	PrecMin	28.1	0.0	0.0	0.0	0.2	1.5	1.6	1.8	3.0	3.6	0.0	0.0	0.0
	PrecStd	6.6	0.6	0.5	1.0	1.7	1.8	1.9	2.6	3.5	2.3	2.2	0.7	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.5	101.9	102.8	102.5	99.5	96.0	91.6	89.7	92.3	96.0	98.4	97.8
		MCWB	62.9	66.9	67.7	72.3	76.6	77.6	77.3	77.4	78.1	73.4	67.6	63.9
	2%	DB	96.3	100.0	101.4	100.5	97.3	93.6	89.6	87.8	90.2	93.9	97.0	95.9
		MCWB	62.4	66.5	68.4	73.5	77.0	77.3	76.8	76.7	77.8	75.4	68.3	63.4
	5%	DB	94.3	98.5	100.1	98.8	95.4	91.7	88.1	86.3	88.6	92.3	95.6	94.3
		MCWB	61.7	65.9	68.7	74.1	77.1	77.1	76.5	76.3	77.4	76.3	69.1	63.0
	10%	DB	92.3	96.6	98.6	97.1	93.6	89.9	86.5	84.8	86.8	90.6	94.2	92.5
		MCWB	60.7	64.6	69.0	74.7	77.3	76.8	76.2	76.0	77.0	76.7	69.9	62.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.1	76.5	78.2	79.2	80.8	80.0	78.7	79.1	80.1	80.6	77.8	75.2
		MCDB	86.2	89.6	91.6	92.9	91.4	88.7	86.6	85.6	87.7	87.9	88.2	88.2
	2%	WB	68.4	75.0	77.1	78.4	79.8	77.8	77.9	77.9	79.0	79.2	76.6	72.7
		MCDB	86.5	90.0	91.3	91.7	90.9	88.8	85.8	84.7	86.6	87.7	87.5	83.5
	5%	WB	64.1	73.5	76.1	77.7	79.1	78.1	77.2	77.2	78.2	78.3	75.7	68.1
		MCDB	89.9	88.3	90.4	91.3	90.1	87.8	85.3	83.9	85.6	87.2	86.7	85.9
	10%	WB	62.0	71.6	75.2	77.0	78.3	77.4	76.7	76.5	77.5	77.6	74.8	65.0
		MCDB	88.6	87.4	89.6	90.9	89.3	86.7	84.4	82.8	84.6	86.4	85.6	86.6
(35) Mean Daily Temperature Range	5% DB	MDBR	30.2	26.9	21.7	17.8	16.4	14.1	13.0	11.9	13.9	18.3	27.1	31.3
		MCDBR	33.4	28.7	24.0	19.9	18.3	16.5	15.2	14.2	16.7	21.0	29.1	33.3
		MCWBR	10.5	9.4	8.1	5.8	5.0	4.6	5.0	5.1	6.2	6.3	7.5	9.1
	5% WB	MCDBR	31.0	21.3	18.6	17.6	16.8	14.8	14.2	13.1	15.7	18.5	22.4	28.8
		MCWBR	11.5	9.3	6.8	5.9	6.0	5.4	5.3	5.5	6.5	7.0	6.5	10.9
(40) Clear-Sky Solar Irradiance	taub		0.598	0.670	0.772	0.675	0.602	0.555	0.503	0.481	0.494	0.557	0.558	0.550
	taud		1.767	1.647	1.511	1.739	1.852	1.978	2.172	2.247	2.180	1.970	1.897	1.888
	Ebn at Noon		229	218	200	219	231	238	252	262	261	243	238	238
	Edh at Noon		71	82	96	75	66	57	48	45	48	59	62	62
(44) All-Sky Solar Radiation	RadAvg		1760	1812	1862	1891	1853	1694	1566	1463	1587	1787	1739	1703
(45)	RadStd		87	81	71	61	74	69	62	59	61	72	47	55

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page